BioMatch – Developer brief · Add-on

Financing check, team model, market model, IP06 & cap table, gap-based value-add, Syndicate Builder, impact-ranked next steps · add-on 1.0 · 24 Oct 2026 · Reference implementation: `addons-config.json` + `addons.js` + `examples.js` (plain JS, no dependencies)

> **Scope.** Extends the core brief v2.0 (question-bank.json + scoring.js). Nothing in the core is replaced: readiness weights, gates G1–G5, checks F1–F10, levels and the matching formulas stay as they are. This add-on adds checks F11–F18, two model-based inputs for TM and MK, question IP06, the Syndicate Builder and match/syndicate impact for next best steps. These carry the pitch: **Wow 2** (coaching incl. "your round does not reach your milestone") and **Wow 3** (Syndicate Builder).

1 · What is added and where it plugs in

Add-on	Plugs into (core)	Function in addons.js
F11 financing check	Step 4 consistency checks	`financingCheck(startup, cfg)`
Team model	Dimension TM (blend), F12, G2 unchanged	`teamModel(startup, cfg)`, `scoreMember(m, cfg)`
Market model	Dimension MK (blend), F13, F18	`marketModel(startup, cfg)`
IP06 + cap table	Dimension IP (new question), F14–F17	`ip06(startup, cfg)`
All new checks	Step 4, run after core F1–F10	`addonChecks(startup, cfg)` → { findings, team, market, ip06 }
Gap-based value-add	Startup fit · Value-add (w 25)	`deriveGaps(findings, coreFiredIds, startup)`, `wantedValueAdd(ticked, gaps, cfg)`
Syndicate Builder	New step 6b after matching	`buildSyndicates(startup, vcs, matches, gaps, cfg, { roundSizeEur })`
Impact for next steps	Next best steps (extends ranking)	`simulateImpact(startup, findings, vcs, core, cfg)` with `core = { readiness, matchAll }`

2 · Entity extensions

Entity	New fields
Startup	`area` (oncology | rare_disease | infectious | cns | immunology | cardiometabolic | other), `country`, `grants[]`, `round{ sizeEur, nextMilestone?, monthsToMilestone, monthlyBurnEur, runwayMonths, leadSecured, reachedMilestoneIndex? }`, `team[TeamMember]`, `market{ patients, annualPriceEur, eligibleSharePct, source, peakSharePct?, firstInClass }`, `ip{ licenceType (assigned|exclusive|non_exclusive|option), priorityYear, institutionEquityPct, founderOwnershipRange (<25|25-50|50-75|>75) }`
TeamMember	name, role (ceo|cso|cto|cbo|coo|cfo|cmo|head_regulatory|head_development|other), founder, fulltime, highestDegree (bsc|msc|phd|md|md_phd|mba|other), businessDegree, institution, domainMatch, hIndex, backgroundYears{ academia, industryRnd, industryCommercial, vcInvesting, consultingBanking, startupOperator }, programsToClinic, dealsClosed, startupsFounded, exits, capitalRaisedEur, seniorRole, peopleLed, selfAssessment{ science, translational, commercial, entrepreneurial, leadership: 0–10 }, evidence[cv|linkedin|reference], verified
VC	`roles[]` must contain "lead" and/or "co"; reserves as `reserveEur` or `reserveRatio × fundSizeEur`; `valueAdd[]` uses the tags in §7
Finding	checkId, severity (critical | warning | open_question | info), dimension, gapTag, title, why, evidence{ summary, … }, actions[{ type, text, … }], impact?
Syndicate	startupId, roundSizeEur, score, shortfallEur, gapCoverage (%), followOnReserveEur, nextRoundEstimateEur, members[{ vcId, name, type, role: lead|co, ticketEur, match, covers[] }], why[]

UI: round block, market inputs and IP06 are new questions (req R from D1 for round + team, from D2 for market + IP06). The team block is a repeatable member form; `selfAssessment` are sliders; CV/LinkedIn upload sets `evidence`.

3 · F11 – Financing check (round vs. milestone)

```
costGroup  = most expensive active module: CT/GT → cgt · BI/RN/PL → bio · SM → sm
milestone  = round.nextMilestone or default by band (D1 development candidate · D2 IND/CTA-enabling · D3 Phase 1 · D4 Phase 2a)
cashOnHand = monthlyBurn × runwayMonths
ownBurn    = monthlyBurn × (monthsToMilestone + 6)        // 6 months to raise the next round
required   = max(mid(cost), ownBurn) ;  available = round.sizeEur + cashOnHand
```

```
available < low(cost) → critical ;  available < required → warning ;  else no finding
targetRound = ceil((required − cashOnHand) / 0.5M) × 0.5M
```

Milestone (EUR M)	sm	bio	cgt
In vivo PoC	0.5–1.5	1–2	1–3
Development candidate	1.5–4	2–5	3–6
IND/CTA-enabling complete	3–8	5–12	8–20
Phase 1 complete	8–20	10–25	15–40
Phase 2a readout	20–50	25–60	30–80

Actions (in this order):

1. Raise to targetRound.
2. Re-scope to the furthest earlier milestone with available ≥ (low+mid)/2, after the milestone already reached (D1 none · D2 in vivo PoC · D3 candidate · D4 IND).
3. Tranches: current amount now, rest on that milestone.
4. Non-dilutive funding not yet used (by country).
5. Syndicate Builder at targetRound.

Cost ranges are demo assumptions in config.

4 · Team model → TM

Each member gets five competencies (0–100) with a "why" line per point. Years use diminishing bands: `≥1y 15 · ≥4y 35 · ≥10y 55 · ≥20y 65`.

Competency	Points
Science	degree (PhD 30, MD/PhD 35, MD 25, MSc 15, BSc 5) + institution (top 20, strong 10; configurable list) + h-index (≥5: 5, ≥15: 10, ≥30: 15) + domain match 15 + 0.4 × band(academia)
Translational	band(industryRnd) + 20 per program to IND/clinic (max 40) + 15 if CMO / head of regulatory or development + 10 if MD
Commercial	max(band(industryCommercial), band(vcInvesting), 0.8 × band(consultingBanking), 0.8 × band(startupOperator)) + 15 MBA/business degree + 10 per deal (max 30)
Entrepreneurial	15 per startup founded (max 30) + 25 per exit (max 50) + capital raised (≥€1M 10, ≥€10M 20, ≥€50M 30)
Leadership	40 previous VP/C-level/director + 0.6 × band(total years) + people led (≥3: 10, ≥10: 20)

```
slider:    competency = max(structured, 10 × selfAssessment)
proof cap: member without CV / LinkedIn / verification → each competency capped at 50
           + proofRequest "Upload CV or LinkedIn for <name> to unlock <n> <competency> points"
team[c]   = best member (part-time × 0.6) + 0.15 × second best, clamp 100
teamScore = Σ stageWeight[band][c] × team[c] − 10 (no CEO / CEO part-time) − 5 (nobody with domain match)
   D1 sci .45 trans .15 com .15 entr .15 lead .10 · D2 .35 .15 .25 .10 .15 · D3 .20 .30 .25 .10 .15 · D4 .15 .30
.25 .10 .20
required ≥ 40: D1 science · D2 + commercial · D3 + translational · D4 + leadership → F12
core integration: s_TM = 0.5 × s_TM(TM01–TM07) + 0.5 × teamScore / 100
```

5 · Market model → MK · IP06

```
TAM = patients × annualPriceEur ;  SAM = TAM × eligibleSharePct ;  peak revenue at 1 % / 5 % / 10 % of SAM
credibleCase = founder peakSharePct if ≤ 30 % (firstInClass) or ≤ 15 % (otherwise), else 5 % base case
PoS = band (D1 .02 · D2 .05 · D3 .07 · D4 .12) × area (oncology .5 · rare 1.3 · infectious 1.5 · CNS .7 · other 1)
riskAdjusted = credibleCase × PoS
marketScore = credibleCase: <€20M 10 · ≥20M 25 · ≥50M 50 · ≥100M 75 · ≥250M 100
core integration: s_MK = 0.5 × s_MK(MK01–MK05) + 0.5 × marketScore / 100   (questions only if inputs missing)

IP06 "Priority year of the core patent" (proof type: patent), required from D2
launchYear = now + (D1 11 · D2 10 · D3 8 · D4 6) years ;  protection = priorityYear + 20 − launchYear
points(IP06) = ≥ 10 y → 1 · 6–9 y → 0.7 · < 6 y → 0.3 · unanswered → 0
```

6 · New consistency checks

ID	Trigger	Severity → effect
F11	Round + cash below cost of the stated milestone (§3)	Critical (below low) / Warning → confirm; coaching card with 5 actions
F12	Required team competency < 40 (§4)	Open question → shown to VCs; action = hire profile
F13	Credible peak revenue < €100M (< €50M → warning)	Info / Warning → coaching (expansion, licensing, strategic investors)
F14	licenceType = option or non_exclusive at ≥ D2	Warning → confirm
F15	IP06 protection at launch < 6 years	Warning → confirm

F16	Institutional equity ≥ 10 % (≥ 20 % → warning)	Info / Warning → confirm
F17	Founder ownership < 25 % (warning) or 25–50 % at D1–D2 (info)	Info / Warning → confirm
F18	Market inputs missing at ≥ D2, no source, or founder share above credible limit	Info → coaching

Warnings fall under gate G4 like the core warnings. Severity "info" never blocks listing.

7 · Gap-based value-add

Startup fit · Value-add (w 25) uses **wanted = value-add ticked by the founder ∪ value-add derived from open gaps**. Gaps come from add-on findings (not info) and from core checks.

Gap tag	Source	Value-add tags that help
commercial_leadership	F12 (commercial / leadership / entrepreneurial), core F9	ceo_recruiting, venture_building
translational	F12 (translational)	cro_network, lab_access, regulatory_expertise
science	F12 (science)	scientific_network, lab_access
ip_licensing	F14, F15, core F3	tto_negotiation, ip_expertise
cap_table	F16, F17	cap_table_restructuring, venture_building
regulatory	core F10	regulatory_expertise
funding_gap	F11, core F6	syndicate_network, grant_expertise
market	F13, F18	pharma_network, market_access
syndication	round.leadSecured = false	syndicate_network

8 · Syndicate Builder (step 6b)

Input: the startup's matches from the core matcher, **computed without the filter "startup needs lead and VC cannot lead"** (co-investors belong in a syndicate). All other hard filters stay.

```
pool  = matches with score ≥ 50
leads = pool with roles ∋ lead, ticket.max ≥ 25 % of round, ticket.min ≤ 60 % of round      // top 5 by score
for each lead:
    leadTicket = clamp(40 % of round, ticket.min, ticket.max)
    while remaining > 0 and members < 5:
        add co-investor maximizing  score + 10 × new gaps covered + 5 × new investor type,
        if ticket.min ≤ min(ticket.max, remaining);  ticket = min(ticket.max, remaining)
    nextRound = mid cost of the milestone after round.nextMilestone (§3)
    synScore  = 0.4 × avg(score) + 30 × gapCoverage + 20 × min(1, Σ reserves / nextRound)
                + 10 × min(1, investor types / 3) − 15 if shortfall
return top 3 distinct member sets with why[] (lead + covered gaps, each co + ticket + gaps, reserves vs next
round, shortfall/uncovered)
```

When F11 fires, run it twice: at round.sizeEur and at targetRound, and show both side by side. "Request intros" creates one IntroRequest per member (optional `syndicateId`). Max 10 open intro requests (core rule) still applies.

9 · Next best steps with impact

```
for each add-on finding with a fix patch:
    F11 → round.sizeEur = targetRound · F12 → add typical hire per missing competency (config.team.typicalHire)
    F14 → licenceType = exclusive · F16 → institutionEquityPct = 8 · F17 → founderOwnershipRange = 50-75
    readinessGain = core.readiness(after) − core.readiness(now)
    matchGain     = avg top-3 core match score (after − now)
    syndicateGain = best syndicate score (after − now)
    rankKey       = readinessGain + 0.5 × matchGain + 0.5 × syndicateGain
merge with core candidates (their rankKey = readinessGain), sort by rankKey desc, quick wins first on ties, show
top 3 + all critical
```

Card text: headline, why investors care, the numbers from `evidence`, actions, one impact line (e.g. "Adding a CBO raises your average match by 6"). An LLM may rephrase the card but never adds or changes numbers.

10 · Test values (examples.js)

examples.js uses stub `core.readiness` / `core.matchAll`; the add-on values below do not depend on them, the impact and syndicate scores do. Replace the stubs with scoring.js.

Case: Lumora (D2, SM, €2M seed)	Expected result

F11	Critical · available €2.45M (2.0 + 0.45 cash) vs cost €3–8M · own burn €2.7M · gap €3.05M · targetRound €5.5M · actions incl. re-scope to development candidate + €3.5M tranche
Team model	Score 35 · science 92, translational 15, commercial 30, entrepreneurial 0, leadership 23 · F12 commercial · proof request: Sara Novak +20 commercial (slider 7, no CV)
Market model	TAM €4.0bn, SAM €2.0bn · 1 % €20M · 5 % €100M · 10 % €200M · founder 40 % not credible → base case €100M · F18 ×2 (share, no source)
IP	Launch ~2036, protection 8 y → IP06 0.7 · F14 (option) · F16 warning (25 %)
Gaps / wanted value-add	funding_gap, commercial_leadership, ip_licensing, cap_table, syndication → + syndicate_network, grant_expertise, ceo_recruiting, venture_building, tto_negotiation, ip_expertise, cap_table_restructuring
Syndicates @ €2.0M	#1 76: Munich Translational (lead €1.0M) + Helix (co €1.0M); reserves €11M vs ~€14M next round
Syndicates @ €5.5M	#1 85: Helix (lead €1.5M) + Public Seed Fund (co €1.0M) + Rhine Pharma CVC (co €3.0M); reserves €30M vs ~€14M